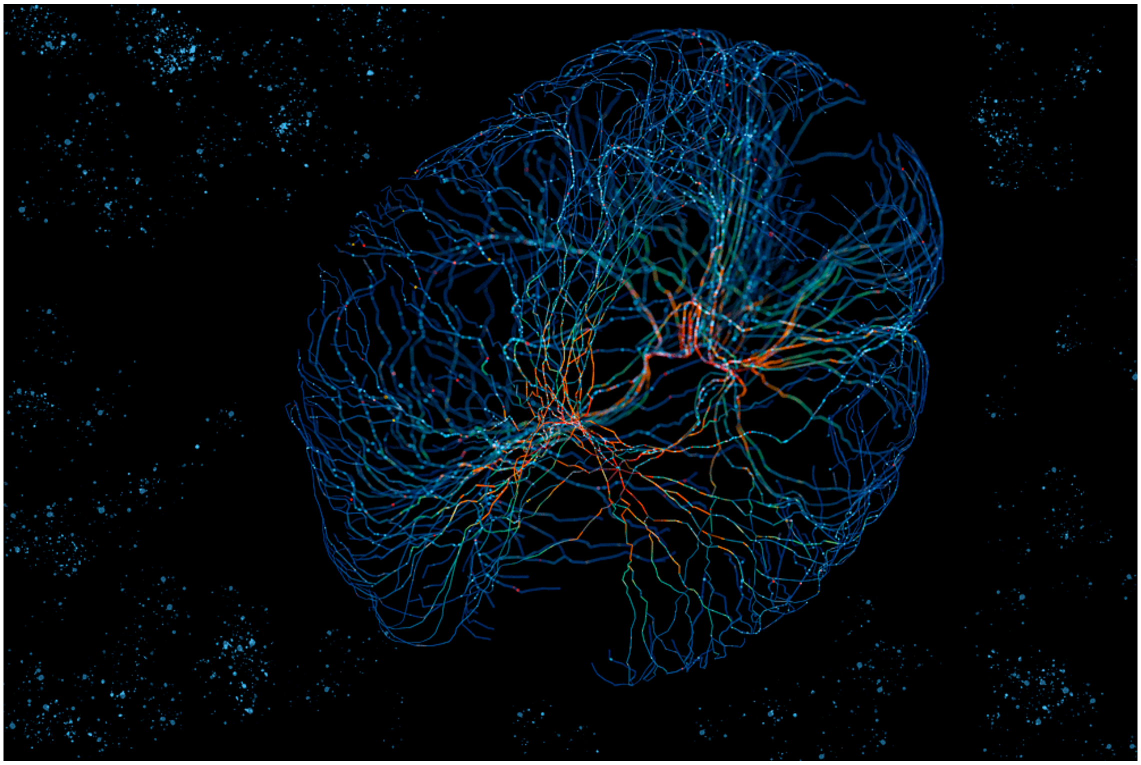
olinkWrapper

Comprehensive Analysis Report



Report Date
March 06, 2026

Software Version
OlinkWrapper v1.5.2

App Developer
Jyotirmoy Das

Disclaimer: Please remember **olinkWrapper** is based on the **OlinkAnalyze** package from the Olink Developers. Please cite/use the original package.

olink-wrapper.serve.scilifelab.se

github.com/JD2112/olinkWrapper

Zenodo: 15098644

Report generated by olinkWrappeR on 2026-03-06 13:17:59.739154

If you're using OlinkWrapper, please cite: Zenodo: 15098644

Contents

1 Introduction

The **olinkWrapper** application is built upon the *OlinkAnalyze* R package (Olink Proteomics 2024b) that builds on the Olink® Proximity Extension Assay (PEA) technology platform - a high-throughput, multiplex immunoassay system enabling simultaneous quantification of up to 96 proteins per panel with high specificity and sensitivity. (Assarsson et al. 2014).

The underlying analytical framework relies on the **OlinkAnalyze** R package developed by Olink Proteomics, which implements statistical workflows for NPX data including normalization, differential expression testing, linear modeling, and pathway enrichment. (Olink Proteomics 2024a).

This report documents all computational steps, statistical analyses, and visualizations performed during the current analytical session.

NOTE:

1. Please check the olinkWrapper manual for more information about how to proceed with the analysis and how to interpret the results.
2. This report **DOES NOT** contain the preprocessing analysis steps that you can download as a separate report.

2 Analysis Workflow

2.1 Step 1: T-Test

Timestamp: 2026-03-06 09:32:07

2.1.1 Background

The olink_ttest function performs a Welch 2-sample t-test or paired t-test at confidence level 0.95 for every protein (by OlinkID) for a given grouping variable. It corrects for multiple testing using the Benjamini-Hochberg method ("fdr"). Adjusted p-values are logically evaluated towards adjusted p-value < 0.05. The resulting t-test table is arranged by ascending p-values.

2.1.2 Execution Parameters

Variable: Treatment | Type: Factor

2.1.3 Results Table (Preview)

NOTE: This is a preview of the results table. The full results table can be downloaded from the app.

Assay	OlinkID	UniProt	Panel	estimate	Treated	Untreated	statistic	p.value	parameter	conf.low	conf.high	method	a
SERPINA7	OID01232	P05543	Olink Cardiometabolic	-3.2079448	9.059112	12.267056	-4.967109	0.0000020	139.8808	-4.4848095	-1.9310801	Welch Two Sample t-test	tw
TRAIL	OID00488	P50591	Olink Inflammation	-2.6337968	7.426476	10.060273	-4.903592	0.0000025	143.5797	-3.6954729	-1.5721207	Welch Two Sample t-test	tw
CXCL11	OID00486	O14625	Olink Inflammation	1.7060979	5.572706	3.866608	4.259658	0.0000364	146.9446	0.9145654	2.4976304	Welch Two Sample t-test	tw
CD6	OID00499	Q8WWJ7	Olink Inflammation	0.8990610	2.735936	1.836875	3.979471	0.0001108	138.8200	0.4523625	1.3457594	Welch Two Sample t-test	tw
Flt3L	OID00533	P49771	Olink Inflammation	1.9919257	6.105514	4.113589	3.919867	0.0001358	146.0072	0.9876233	2.9962281	Welch Two Sample t-test	tw
MMP-10	OID00527	P09238	Olink Inflammation	-2.0610224	9.279465	11.340488	-3.881539	0.0001629	132.5349	-3.1113156	-1.0107292	Welch Two Sample t-test	tw
DPP4	OID01266	P27487	Olink Cardiometabolic	-1.9558861	4.234631	6.190517	-3.692686	0.0003094	151.0851	-3.0023928	-0.9093794	Welch Two Sample t-test	tw
TWEAK	OID00555	O43508	Olink Inflammation	-1.9010203	8.132830	10.033850	-3.510039	0.0005903	151.5630	-2.9710714	-0.8309693	Welch Two Sample t-test	tw
DEFA1	OID01277	P59665	Olink Cardiometabolic	0.8790020	4.313249	3.434247	3.318406	0.0011424	146.3701	0.3555052	1.4024987	Welch Two Sample t-test	tw
EFEMP1	OID01281	Q12805	Olink Cardiometabolic	-0.6785813	2.089538	2.768119	-3.149604	0.0019668	152.9101	-1.1042238	-0.2529389	Welch Two Sample t-test	tw
IL-22 RA1	OID00516	Q8N6P7	Olink Inflammation	-2.5925596	9.021438	11.613997	-3.082687	0.0024366	151.9843	-4.2541320	-0.9309871	Welch Two Sample t-test	tw
REG3A	OID01280	Q06141	Olink Cardiometabolic	-2.1991838	7.226846	9.426030	-3.060666	0.0026416	141.9620	-3.6195872	-0.7787804	Welch Two Sample t-test	tw
ICAM1	OID01230	P05362	Olink Cardiometabolic	-0.5667395	2.497806	3.064546	-3.009358	0.0031268	134.4697	-0.9392031	-0.1942759	Welch Two Sample t-test	tw
TCN2	OID01259	P20062	Olink Cardiometabolic	-2.8578636	5.544592	8.402456	-2.909307	0.0042546	131.4817	-4.8010562	-0.9146710	Welch Two Sample t-test	tw
COL18A1	OID01271	P39060	Olink Cardiometabolic	-1.9710734	7.749768	9.720842	-2.829081	0.0053057	150.3726	-3.3476955	-0.5944513	Welch Two Sample t-test	tw
LILRB5	OID01220	O75023	Olink Cardiometabolic	1.2949857	4.714865	3.419879	2.827611	0.0053237	151.5512	0.3901383	2.1998330	Welch Two Sample t-test	tw
TNFSF14	OID00506	O43557	Olink Inflammation	-1.6872463	4.419953	6.107199	-2.782720	0.0061737	133.2836	-2.8865198	-0.4879728	Welch Two Sample t-test	tw
SAA4	OID01269	P35542	Olink Cardiometabolic	1.9055236	9.771593	7.866069	2.755773	0.0066402	139.1026	0.5383810	3.2726662	Welch Two Sample t-test	tw
C2	OID01233	P06681	Olink Cardiometabolic	1.5719926	9.301200	7.729207	2.733058	0.0070388	148.2208	0.4353862	2.7085990	Welch Two Sample t-test	tw
OPG	OID00479	O00300	Olink Inflammation	-0.6567817	3.351215	4.007996	-2.708813	0.0075549	146.8992	-1.1359440	-0.1776194	Welch Two Sample t-test	tw
IL-15RA	OID00514	Q13261	Olink Inflammation	-0.6422560	2.687487	3.329743	-2.641458	0.0091201	151.6360	-1.1226445	-0.1618675	Welch Two Sample t-test	tw
VCAM1	OID01257	P19320	Olink Cardiometabolic	1.0233463	11.039989	10.016642	2.585504	0.0106691	151.0649	0.2413250	1.8053676	Welch Two Sample t-test	tw
CA1	OID01223	P00915	Olink Cardiometabolic	-1.0648537	3.255181	4.320035	-2.575405	0.0109615	152.6889	-1.8817150	-0.2479924	Welch Two Sample t-test	tw
CD5	OID00531	P06127	Olink Inflammation	1.4115515	9.238005	7.826453	2.494960	0.0136685	151.6625	0.2937607	2.5293424	Welch Two Sample t-test	tw
IL7R	OID01253	P16871	Olink Cardiometabolic	-0.7078341	2.694697	3.402531	-2.444350	0.0156479	152.9352	-1.2799268	-0.1357413	Welch Two Sample t-test	tw

2.2 Step 2: Pathway Enrichment

Timestamp: 2026-03-06 09:38:25

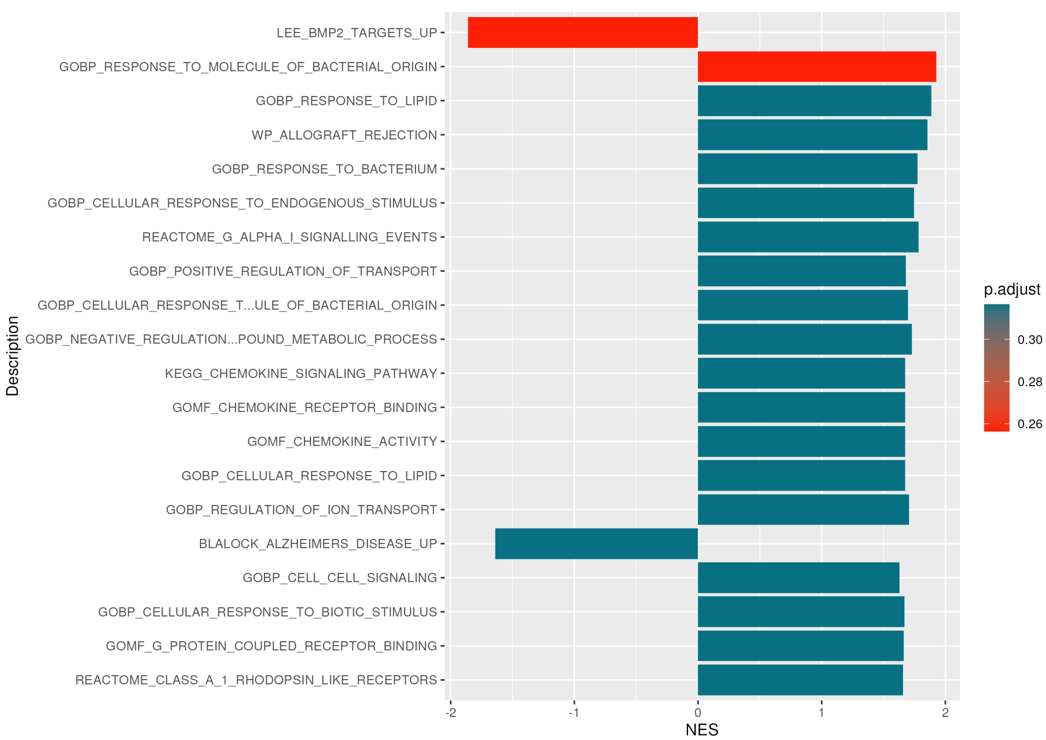
2.2.1 Background

The olink_pathway_enrichment function can be used to perform Gene Set Enrichment Analysis (GSEA) or Over-representation Analysis (ORA) using MSigDB, Reactome, KEGG, or GO. MSigDB includes curated gene sets (C2) and ontology gene sets (C5) which encompasses Reactome, KEGG, and GO. This function performs enrichment using the gsea or enrich functions from clusterProfiler from BioConductor. The function uses the estimate from a previous statistical analysis for one contrast for all proteins. MSigDB is subset if ontology is KEGG, GO, or Reactome. test_results must contain estimates for all assays. Posthoc results can be used but should be filtered for one contrast to improve interpretability. Alternative statistical results can be used as input as long as they include the columns “OlinkID”, “Assay”, and “estimate”. A column named “Adjusted_pal” is also needed for ORA. Any statistical results that contains one estimate per protein will work as long as the estimates are comparable to each other.

2.2.2 Execution Parameters

Method: GSEA | Ontology: MSigDb | Organism: human

2.2.3 Visualization



2.2.4 Results Table (Preview)

NOTE: This is a preview of the results table. The full results table can be downloaded from the app.

	ID
GOBP_RESPONSE_TO_MOLECULE_OF_BACTERIAL_ORIGIN	GOBP_RESPONSE_TO_MOLECULE_OF_BACTERIAL_ORIGIN
LEE_BMP2_TARGETS_UP	LEE_BMP2_TARGETS_UP
GOBP_RESPONSE_TO_LIPID	GOBP_RESPONSE_TO_LIPID
WP_ALLOGRAFT_REJECTION	WP_ALLOGRAFT_REJECTION
REACTOME_G_ALPHA_I_SIGNALLING_EVENTS	REACTOME_G_ALPHA_I_SIGNALLING_EVENTS
GOBP_RESPONSE_TO_BACTERIUM	GOBP_RESPONSE_TO_BACTERIUM
GOBP_CELLULAR_RESPONSE_TO_ENDOGENOUS_STIMULUS	GOBP_CELLULAR_RESPONSE_TO_ENDOGENOUS_STIMULUS
GOBP_NEGATIVE_REGULATION_OF_NUCLEOBASE_CONTAINING_COMPOUND_METABOLIC_PROCESS	GOBP_NEGATIVE_REGULATION_OF_NUCLEOBASE_CONTAINING_COMPOUN
GOBP_REGULATION_OF_ION_TRANSPORT	GOBP_REGULATION_OF_ION_TRANSPORT
GOBP_CELLULAR_RESPONSE_TO_MOLECULE_OF_BACTERIAL_ORIGIN	GOBP_CELLULAR_RESPONSE_TO_MOLECULE_OF_BACTERIAL_ORIGIN
GOBP_POSITIVE_REGULATION_OF_TRANSPORT	GOBP_POSITIVE_REGULATION_OF_TRANSPORT
GOBP_CELLULAR_RESPONSE_TO_LIPID	GOBP_CELLULAR_RESPONSE_TO_LIPID
GOMF_CHEMOKINE_ACTIVITY	GOMF_CHEMOKINE_ACTIVITY
GOMF_CHEMOKINE_RECEPTOR_BINDING	GOMF_CHEMOKINE_RECEPTOR_BINDING
KEGG_CHEMOKINE_SIGNALING_PATHWAY	KEGG_CHEMOKINE_SIGNALING_PATHWAY
GOBP_CELLULAR_RESPONSE_TO_BIOTIC_STIMULUS	GOBP_CELLULAR_RESPONSE_TO_BIOTIC_STIMULUS
GOMF_G_PROTEIN_COUPLED_RECEPTOR_BINDING	GOMF_G_PROTEIN_COUPLED_RECEPTOR_BINDING
REACTOME_CLASS_A_1_RHODOPSIN_LIKE_RECEPTORS	REACTOME_CLASS_A_1_RHODOPSIN_LIKE_RECEPTORS
REACTOME_GPCR_LIGAND_BINDING	REACTOME_GPCR_LIGAND_BINDING
REACTOME_PEPTIDE_LIGAND_BINDING_RECEPTORS	REACTOME_PEPTIDE_LIGAND_BINDING_RECEPTORS
REACTOME_SIGNALING_BY_GPCR	REACTOME_SIGNALING_BY_GPCR
BLALOCK_ALZHEIMERS_DISEASE_UP	BLALOCK_ALZHEIMERS_DISEASE_UP
GOBP_CELL_CELL_SIGNALING	GOBP_CELL_CELL_SIGNALING
BLALOCK_ALZHEIMERS_DISEASE_INCIPIENT_UP	BLALOCK_ALZHEIMERS_DISEASE_INCIPIENT_UP
GOBP_RESPONSE_TO_CHEMOKINE	GOBP_RESPONSE_TO_CHEMOKINE

2.3 Step 3: Descriptive Stats

Timestamp: 2026-03-06 13:00:56

2.3.1 Background

Provides a global statistical summary of the dataset, including mean, median, quartiles, and range for all variables. It helps in understanding the overall data characteristics.

2.3.2 Execution Parameters

Global summary statistics computed per assay (Mean, Median, SD, etc.).

2.3.3 Results Table (Preview)

NOTE: This is a preview of the results table. The full results table can be downloaded from the app.

Assay	n	Mean	Median	SD	Min	Max	Missing
4E-BP1	158	2.461	2.479	1.279	-0.702	5.240	0
ADA	158	4.743	4.833	2.369	-1.783	9.953	0
ANG	159	3.483	3.272	1.913	-0.596	9.564	0
ANGPTL3	159	9.369	8.698	4.204	-1.899	21.643	0
AOC3	159	6.469	6.437	3.454	-1.921	14.950	0
APOM	159	10.224	10.257	3.939	-0.017	20.146	0
ARTN	158	3.443	3.408	1.917	-3.429	8.246	0
AXIN1	158	7.904	7.938	3.884	-4.030	19.403	0
Beta-NGF	158	9.754	9.571	4.991	-3.358	22.043	0
C1QTNF1	159	3.428	3.442	2.184	-4.696	8.015	0
C2	159	8.453	8.374	3.608	-0.977	17.316	0
CA1	159	3.837	3.803	2.644	-1.581	10.530	0
CA3	159	9.200	8.699	5.539	-7.466	23.070	0
CA4	159	6.165	6.206	2.279	0.677	12.386	0
CASP-8	158	6.863	6.931	3.258	-1.419	13.833	0
CCL11	158	4.544	4.421	2.282	-0.546	10.786	0
CCL14	159	10.050	10.169	4.054	-1.246	21.008	0
CCL18	159	7.596	7.153	4.631	-4.291	18.760	0
CCL19	158	2.830	2.834	1.200	-0.252	6.435	0
CCL20	158	1.474	1.506	0.839	-0.308	3.787	0
CCL23	158	3.670	3.624	2.585	-3.534	11.221	0
CCL25	158	9.992	10.208	4.122	1.619	20.649	0
CCL28	158	9.847	9.844	3.513	-1.348	21.242	0
CCL3	158	1.201	1.172	0.710	-0.179	3.506	0
CCL4	158	4.860	4.837	2.226	-0.279	10.439	0

2.4 Step 4: Volcano Plot

Timestamp: 2026-03-06 13:17:41

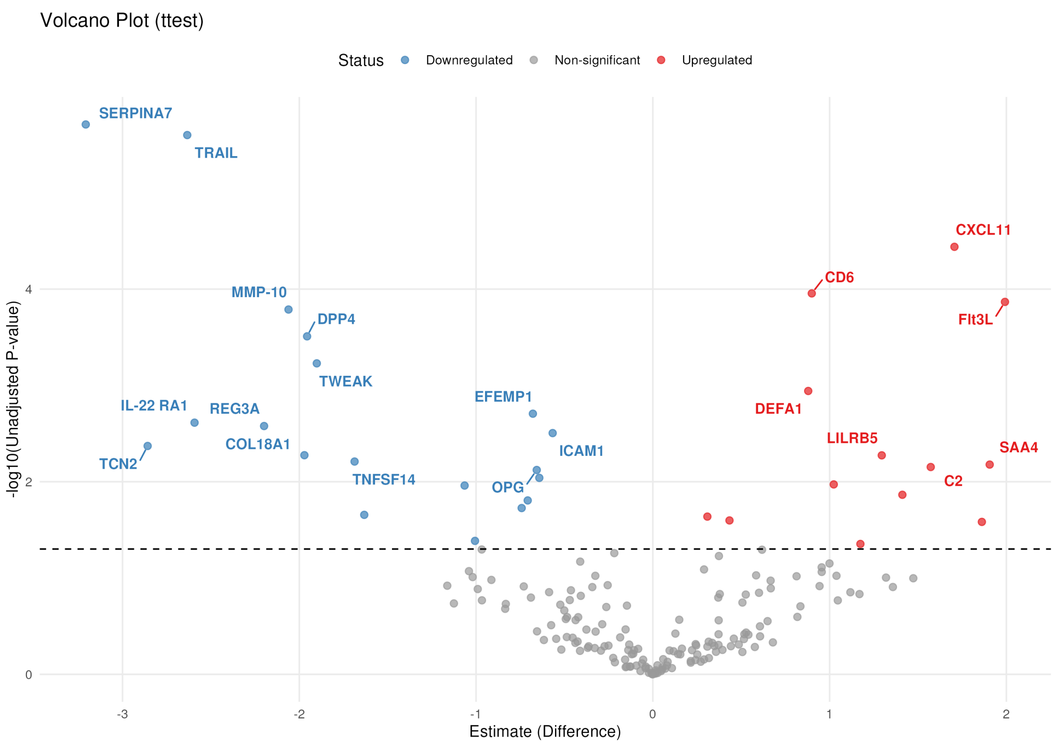
2.4.1 Background

The olink_volcano_plot function generates a volcano plot using results from the olink_ttest function. The estimated difference is shown in the x-axis and -log10(p-value) in the y-axis. The horizontal dotted line indicates p-value = 0.05. Dots are colored based on significance following Benjamini-Hochberg adjustment with a p-value cutoff of 0.05. Significant assays after adjustment can optionally be annotated by OlinkID.

2.4.2 Execution Parameters

Method: ttest | P-val Type: p.value | P-threshold: 0.05

2.4.3 Visualization



2.5 Step 5: Pathway Heatmap

Timestamp: 2026-03-06 13:17:55

2.5.1 Background

The olink_pathway_heatmap function generates a heatmap of proteins related to pathways using the enrichment results from the olink_pathway_enrichment function. Either the top terms can be visualized or terms containing a certain keyword. For each term, the proteins in the test_result data frame that are related to that term will be visualized by their estimate. This visualization can be used to determining how many proteins of interest are involved in a particular pathway and in which direction their estimates are.

2.5.2 Execution Parameters

Method: GSEA | Keyword:

2.5.3 Visualization

3 Session Information

```
## R version 4.4.1 (2024-06-14)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 22.04.5 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;  LAPACK version 3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] digest_0.6.37          kableExtra_1.4.0       knitr_1.48
##  [4] clusterProfiler_4.14.6 umap_0.2.10.0          patchwork_1.3.0
##  [7] ggrepel_0.9.6          broom_1.0.12           DT_0.33
## [10] shinyjs_2.1.0          lmerTest_3.1-3         lme4_1.1-36
## [13] Matrix_1.7-1           writexl_1.5.4          viridis_0.6.5
## [16] viridisLite_0.4.2      plotly_4.10.4          msigdbr_7.5.1
## [19] lubridate_1.9.3        forcats_1.0.0          stringr_1.5.1
## [22] dplyr_1.1.4            purrr_1.2.1            readr_2.1.5
## [25] tidyr_1.3.1            tibble_3.2.1           ggplot2_4.0.2
## [28] tidyverse_2.0.0        OlinkAnalyze_4.0.2     bslib_0.8.0
## [31] shiny_1.9.1
##
## loaded via a namespace (and not attached):
##   [1] splines_4.4.1           later_1.3.2             ggplotify_0.1.2
##   [4] R.oo_1.26.0             cellranger_1.1.0        lifecycle_1.0.4
##   [7] Rdpack_2.6.2            rstatix_0.7.2           lattice_0.22-6
##  [10] vroom_1.6.5             MASS_7.3-61             crosstalk_1.2.1
##  [13] backports_1.5.0         magrittr_2.0.3          sass_0.4.9
##  [16] rmarkdown_2.28          jquerylib_0.1.4         yaml_2.3.10
##  [19] httpuv_1.6.15           ggtangle_0.0.7          askpass_1.2.1
##  [22] reticulate_1.39.0       cowplot_1.2.0           DBI_1.2.3
##  [25] minqa_1.2.8             RColorBrewer_1.1-3      abind_1.4-8
##  [28] zlibbioc_1.52.0         R.utils_2.12.3          BiocGenerics_0.52.0
##  [31] yulab.utils_0.1.7       GenomeInfoDbData_1.2.13 IRanges_2.40.1
##  [34] S4Vectors_0.44.0        enrichplot_1.26.6       tidytree_0.4.6
##  [37] RSpectra_0.16-2         svglite_2.2.2           codetools_0.2-20
##  [40] xml2_1.3.6              DOSE_4.0.1              tidyselect_1.2.1
##  [43] aplot_0.2.3             UCSC.utils_1.2.0        farver_2.1.2
##  [46] stats4_4.4.1            jsonlite_1.8.9          Formula_1.2-5
```

```
##  [49] emmeans_1.10.5         systemfonts_1.3.1       tools_4.4.1
##  [52] treeio_1.30.0          ragg_1.3.3              Rcpp_1.0.13
##  [55] glue_1.8.0             gridExtra_2.3           xfun_0.48
##  [58] qvalue_2.38.0          GenomeInfoDb_1.42.3     withr_3.0.2
##  [61] numDeriv_2016.8-1.1    fastmap_1.2.0           boot_1.3-31
##  [64] fansi_1.0.6            openssl_2.2.2           timechange_0.3.0
##  [67] R6_2.5.1               mime_0.12               gridGraphics_0.5-1
##  [70] estimability_1.5.1     textshaping_0.4.0       colorspace_2.1-1
##  [73] GO.db_3.20.0           RSQLite_2.3.7           R.methodsS3_1.8.2
##  [76] utf8_1.2.4             generics_0.1.3          data.table_1.16.2
##  [79] httr_1.4.7             htmlwidgets_1.6.4       pkgconfig_2.0.3
##  [82] gtable_0.3.6           blob_1.2.4              S7_0.2.1
##  [85] XVector_0.46.0         htmltools_0.5.8.1       carData_3.0-5
##  [88] fgsea_1.32.4           scales_1.4.0            Biobase_2.66.0
##  [91] png_0.1-8              reformulas_0.4.0        ggfun_0.1.7
##  [94] rstudioapi_0.17.1      tzdb_0.4.0              reshape2_1.4.4
##  [97] nlme_3.1-166           nloptr_2.1.1            cachem_1.1.0
## [100] parallel_4.4.1         AnnotationDbi_1.68.0    pillar_1.9.0
## [103] grid_4.4.1             vctrs_0.6.5             promises_1.3.0
## [106] car_3.1-3              xtable_1.8-4            evaluate_1.0.1
## [109] mvtnorm_1.3-1          cli_3.6.3               compiler_4.4.1
## [112] rlang_1.1.4            crayon_1.5.3            labeling_0.4.3
## [115] plyr_1.8.9             fs_1.6.4                stringi_1.8.4
## [118] BiocParallel_1.40.2    babelgene_22.9          Biostrings_2.74.1
## [121] lazyeval_0.2.2         GOSemSim_2.32.0         hms_1.1.3
## [124] bit64_4.5.2            KEGGREST_1.46.0         rbibutils_2.3
## [127] fontawesome_0.5.2      igraph_2.1.1            memoise_2.0.1
## [130] ggtree_3.14.0          fastmatch_1.1-4         bit_4.5.0
## [133] readxl_1.4.3           ape_5.8                 gson_0.1.0
```

4 References

Assarsson, Erika et al. 2014. "A Single-Tube, Quantitative Technique for High-Throughput Protein Analysis." *Nature Methods* 11 (6): 665–70. https://doi.org/10.1038/nmeth.2935.

Olink Proteomics. 2024a. "Olink Proteomics Official Website." https://www.olink.com/.

———. 2024b. "OlinkAnalyze." https://cran.r-project.org/web/packages/OlinkAnalyze/index.html.